Software Engineering

Lecture 4

(Design Principle)

Working on legacy code

- ❑ Re-reading code multiple times to get to the part you need to change.
- ❑ Hard to understand what a method does.
- ❑ Spending a lot of time to fix a minor bug.
- ❑ **You spend more time reading than writing code.**

Principles of Software Engineering

- By the definition, **software engineering** is known as a systematic and procedural approach to software development.

- There are some **basic principles** governing **good** software engineering, of which three popular ones are described below:

  – **DRY** (*"Don't Repeat Yourself"*)

  – **KISS** (*"Keep It Short and Simple/Stupid"*)

  – **SOLID** design principles

Principles of Software Engineering

DRY (*"Don't Repeat Yourself"*)

- The ***"Don't Repeat Yourself"*** (or ***"Duplication Is Evil")*** principle tells us that
  - Every software engineer should aim to reduce repetition of information or methods within their work in order to avoid redundancy. expendability
  - Therefore, it's **recommended** to segregate the entire system of consideration into **fragments**.
  - Dividing the code into smaller segments can help manage the code and use a single segment at any point, by calling, whenever required.
- **DRY Benefits**
  - Less code is good: It saves time and effort, is easy to maintain, and also reduces the chances of bugs.

Principles of Software Engineering

KISS (*"Keep It Short and Simple/Stupid"*)

- The ***"Keep It Short and Simple/Stupid"*** principle reminds us that
  - All software or applications design and deployment should be done as simple as possible with least complexities and clear to understand development procedures.
  - This principle ensures that the source code is made easy to debug (whenever required) and future maintenance by any other operations and maintenance software engineer becomes easier.
  - **Each method** should only solve one small problem, not many use cases.
  - After all, programming languages are for humans to understand.
- **KISS Benefits**
  - It will not only be easier to read and maintain, but it can help find bugs a lot faster.

Working on a startup product

- In charge of the development process
- **Constantly** adding new features
- No formal process
- Very dynamic environment, no time to worry about code structure
- **What is like to go back to your code after 2 years?**

The purpose of SOLID design principles

- To make the code **more maintainable**.
- To make it **easier to quickly extend** the system with new functionality **without breaking the existing ones**.
- To make the code **easier to read and understand**, thus spend less time figuring out what it does and more time actually developing the solution.
- Introduced by **Robert Martin (Uncle Bob)**, named by Michael Feathers.

The purpose of SOLID design principles

- In general, SOLID helps us manage code complexity. It leads to more maintainable and extensible code. Even with big change requests, it's easier to update the code.

- **SOLID** is really a guideline and not a rule.

SOLID design principles

- The SOLID Principles are five principles of Object-Oriented class design.
- They are a set of rules and best practices to follow while designing a class structure.
- These five principles help us understand the need for certain design patterns and software architecture in general.
- It becomes easier for you to develop software that can be managed easily. The other features of using S.O.L.I.D are:
  - It avoids code smells.
  - Quickly refractor code.
  - Can do adaptive or agile software development.

What is the meaning of S.O.L.I.D?

- Following the SOLID acronym, they are:
  - The **S**ingle Responsibility Principle
  - The **O**pen-Closed Principle
  - The **L**iskov Substitution Principle
  - The **I**nterface Segregation Principle
  - The **D**ependency Inversion Principle

Single Responsibility Principle (SRP)

- A class **should have one, and only one**, reason to change.
- A class should only be responsible for one thing.
- There's a place for everything is in its place.
- Find one reason to change and take everything else out of the class.
- Very precise names for many small classes > generic names for large classes.

Single Responsibility Principle (SRP)

```java
class Book {

    String title;
    String author;

    String getTitle() {
        return title;
    }
    void setTitle(String title) {
        this.title = title;
    }
    String getAuthor() {
        return author;
    }
    void setAuthor(String author) {
        this.author = author;
    }
    void searchBook() {...}

}
```

- This code violates the SRP, as the Book class has two responsibilities. **First**, it sets the data related to the books (title and author). **Second**, it searches for the book in the inventory.
- The **setter** methods change the Book object, which might cause problems when we want to search the same book in the inventory.

To apply the Single Responsibility Principle, we need to decouple the two responsibilities

Single Responsibility Principle (SRP)

Before refactoring

```
class Book {

    String title;
    String author;

    String getTitle() {
        return title;
    }
    void setTitle(String title) {
        this.title = title;
    }
    String getAuthor() {
        return author;
    }
    void setAuthor(String author) {
        this.author = author;
    }
    void searchBook() {...}

}
```

- In the refactored code, the **Book** class will only be responsible for getting and setting the data of the Book object.
- **InventoryView** class is only responsible for checking/searching the inventory.

After refactoring

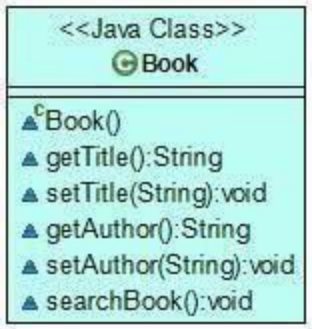
```
class Book {

    String title;
    String author;

    String getTitle() {
        return title;
    }
    void setTitle(String title) {
        this.title = title;
    }
    String getAuthor() {
        return author;
    }
    void setAuthor(String author) {
        this.author = author;
    }

}
```

```
class InventoryView {

  Book book;

    InventoryView(Book book) {
        this.book = book;
    }

    void searchBook() {...}

}
```

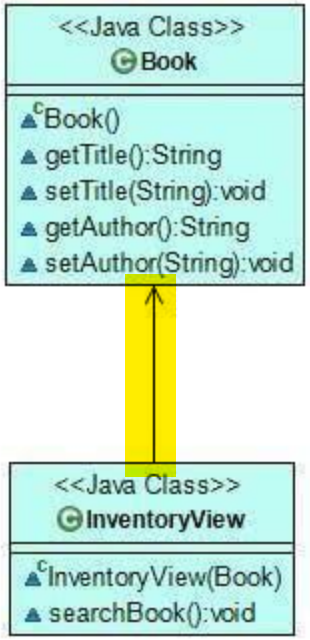
const

Single Responsibility Principle (SRP)

- On the UML diagram below, we can see how the architecture changed after we refactored the code following the Single Responsibility Principle. We split the initial Book class that had two responsibilities into two classes, each having its own single responsibility.



Before refactoring



After refactoring

Single Responsibility Principle (SRP)

- When SRP is followed, **testing is easier**. With a single responsibility, the class will have fewer test cases.
- Less functionality also means **less dependencies** to other modules or classes.
- It leads to better code organization since smaller and well-purposed classes are easier to search.

Open/Closed Principle

- The **Open/Closed Principle** states that classes, modules, microservices, and other code units should be open for extension but closed for modification.
- So, we should be able to extend the existing code using OOP features like inheritance via subclasses and interfaces. However, we should never modify classes, interfaces, and other code units that already exist (**especially if those codes are in production already**), as it can lead to unexpected behavior.
- If we add a new feature by extending our code rather than modifying it, it will minimize the risk of failure as much as possible. Besides, we also don't have to unit test existing functionalities.

Open/Closed Principle

```
class CookbookDiscount {

    String getCookbookDiscount() {

        String discount = "30% between Dec 1 and 24.";

        return discount;
    }

}

class DiscountManager {

    void processCookbookDiscount(CookbookDiscount discount) {...}

}
```

- **CookbookDiscount** to hold the details of the discount.
- **DiscountManager** to apply the discount to the price.

Open/Closed Principle

```
class BiographyDiscount {

    String getBiographyDiscount() {

        String discount = "50% on the subject's birthday.";

        return discount;

    }
}
```

To add the new feature, for example, every biography with a 50% discount on the subject's birthday, we create a new **BiographyDiscount** class.

To process the new type of discount, we need to add the new functionality to the **DiscountManager** class, too

```
class DiscountManager {

    void processCookbookDiscount(CookbookDiscount discount) {...}

    void processBiographyDiscount(BiographyDiscount discount) {...}

}
```

However, as we changed existing functionality, we violated the Open/Closed Principle.

Open/Closed Principle

```java
public interface BookDiscount {

    String getBookDiscount();

}

class CookbookDiscount implements BookDiscount {

    @Override
    public String getBookDiscount() {
        String discount = "30% between Dec 1 and 24.";

        return discount;
    }

}

class BiographyDiscount implements BookDiscount {

    @Override
    public String getBookDiscount() {
        String discount = "50% on the subject's birthday.";

        return discount;
    }

}
```

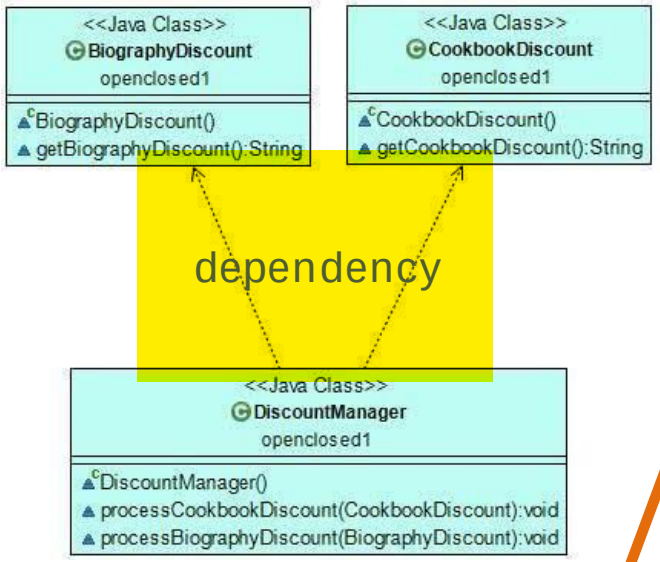
- We need to **refactor** the code by adding an extra layer of abstraction that represents all types of discounts.
- Let's create a new interface called **BookDiscount** that the **CookbookDiscount** and **BiographyDiscount** classes will implement.

```java
class DiscountManager {

    void processBookDiscount(BookDiscount discount) {...}
}
```

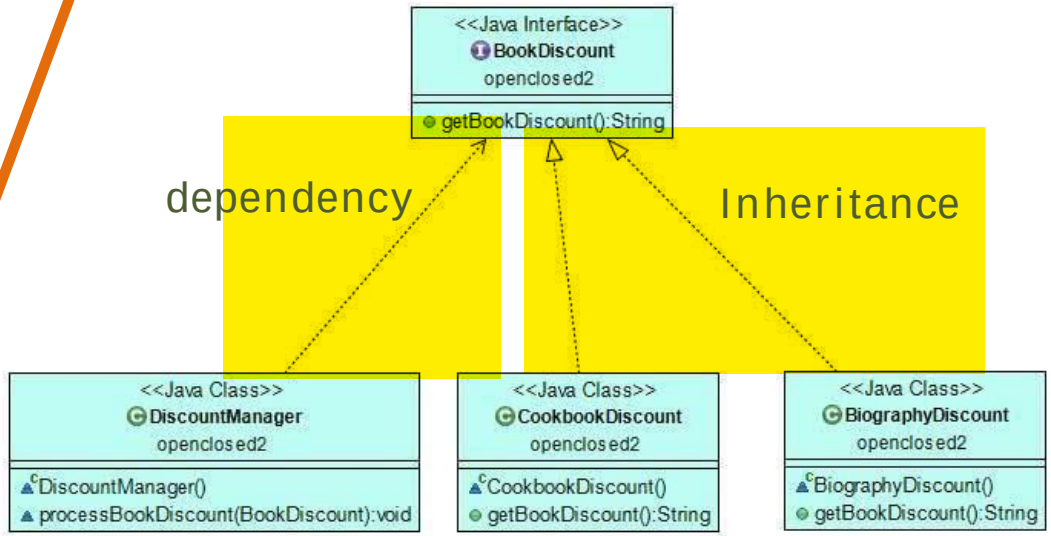
Now, **DiscountManager** can refer to the **BookDiscount** interface instead of the **concrete** classes.

Open/Closed Principle

Before refactoring



After refactoring



Open/Closed Principle

- An entity should be open for extension but closed for modification.
- Extend functionality by adding new code instead of changing the existing code.
- Separate the behaviors, so the system can easily be extended, but never broken.
- **Goal: get a point where you can never break the core of your system.**

Open/Closed Principle

- What this means in essence is that you should design your classes and modules with possible future updates in mind, so they should have a generic design that you won't need to change the class itself in order to extend their behavior.

- You can add more fields or methods, but in such a way that you don't need to rewrite old methods, delete old fields and modify the old code in order to make it work again. Thinking ahead will help you write stable code, before and after an update of requirements.

Liskov Substitution Principle

- *"In a computer program, if **S** is a subtype of **T**, then objects of type T **may** be replaced with objects of type S (i.e., objects of type S may substitute objects of type T) without altering any of the desirable properties of that program (correctness, task performed, etc.)."*

- In layman's terms, it states that an object of a superclass should be replaceable by objects of its subclasses without causing issues in the application.

- We can implement the **Liskov Substitution Principle** by paying attention to the **correct** inheritance hierarchy.

Liskov Substitution Principle

A **BookDelivery** class that informs customers about the number of locations where they can collect their order.

```
class BookDelivery {
    String titles;
    int userID;

    void getDeliveryLocations() {...}
}
```

The store also sells fancy hardcovers they only want to deliver to their high street shops.

```
class HardcoverDelivery extends BookDelivery {

    @Override
    void getDeliveryLocations() {...}

}
```

Later, the store asks us to create delivery functionalities for audiobooks, too.

```
class AudiobookDelivery extends BookDelivery {

    @Override
    void getDeliveryLocations() {...}
}
```

Can't be implemented

Liskov Substitution Principle

To solve the problem, we need to fix the inheritance hierarchy by adding an extra layer that better differentiates book delivery types.

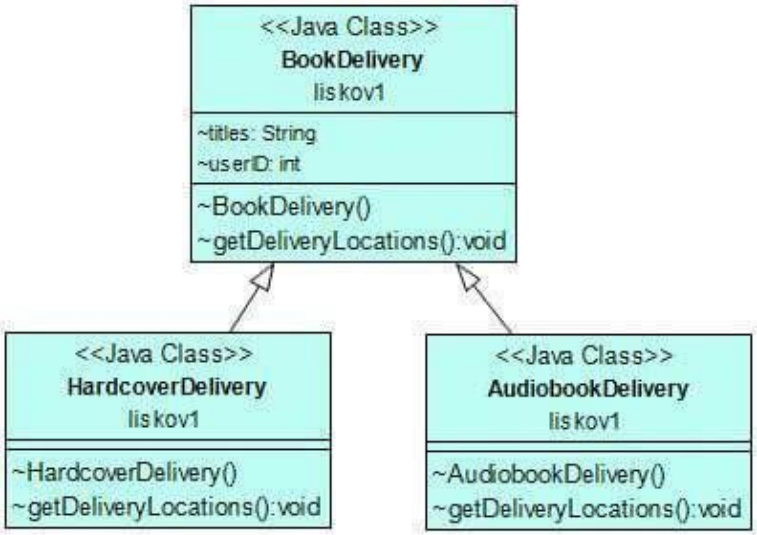
```
class BookDelivery {

    String title;
    int userID;

}

class OfflineDelivery extends BookDelivery {

    void getDeliveryLocations() {...}

}

class OnlineDelivery extends BookDelivery {

    void getSoftwareOptions() {...}

}
```

After refactoring

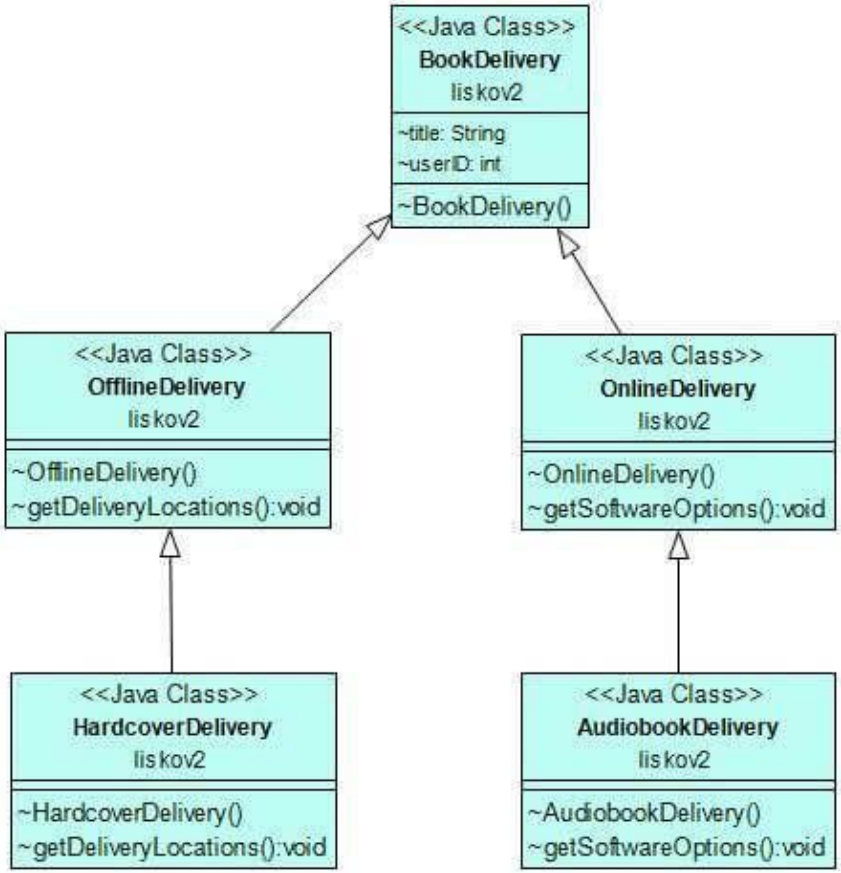
```
class HardcoverDelivery extends OfflineDelivery {

    @Override
    void getDeliveryLocations() {...}

}

class AudiobookDelivery extends OnlineDelivery {

    @Override
    void getSoftwareOptions() {...}

}
```

Liskov Substitution Principle

Before refactoring



After refactoring



Liskov Substitution Principle

- Any derived class should be able to substitute its parent class without the consumer knowing it.
- Every class that implements an interface, must be able to substitute any reference throughout the code that implements that same interface.
- Every part of the code should get the expected result no matter what instance of a class you send to it, given it implements the same interface.

Interface Segregation Principle

- **Classes** (that implement interfaces) should not be forced to depend on methods they don't use. In other words, **interfaces** shouldn't include **too many** functionalities.

- The **violation** of Interface Segregation Principle harms code readability and **forces** programmers to write **dummy methods** that do nothing.

- In a well-designed application, we should avoid interface pollution (also called **fat interfaces**). The **solution** is to create **smaller interfaces** that can be implemented more flexibly.

Interface Segregation Principle

Let's add some user actions to our online bookstore so that customers can interact with the content before making a purchase. To do so, we **create an interface** called ***BookAction*** with three methods: ***seeReviews()***, ***searchSecondHand(),*** and ***listenSample()***.

```
public interface BookAction {

    void seeReviews();
    void searchSecondhand();
    void listenSample();

}
```

Then, we create two classes: **HardcoverUI** and an **AudiobookUI** that implement the ***BookAction*** interface with their own functionalities:

```
class HardcoverUI implements BookAction {

    @Override
    public void seeReviews() {...}

    @Override
    public void searchSecondhand() {...}

    @Override
    public void listenSample() {...}

}

class AudiobookUI implements BookAction {

    @Override
    public void seeReviews() {...}

    @Override
    public void searchSecondhand() {...}

    @Override
    public void listenSample() {...}

}
```

- Both **HardcoverUI** and an **AudiobookUI** depend on methods **they don't use**, so Interface Segregation Principle is **violated**. Hardcover books can't be listened to, so the **HardcoverUI** class doesn't need the listenSample() method. Similarly, audiobooks don't have second-hand copies, so the **AudiobookUI** class doesn't need it, either.
- Thus, ***BookAction*** is a polluted interface that we need to segregate.

Interface Segregation Principle

Solution is to extend ***BookAction*** interface with two more specific sub-interfaces: ***HardcoverAction*** and ***AudioAction***.

```java
public interface BookAction {

    void seeReviews();

}

public interface HardcoverAction extends BookAction {

    void searchSecondhand();

}

public interface AudioAction extends BookAction {

    void listenSample();

}
```

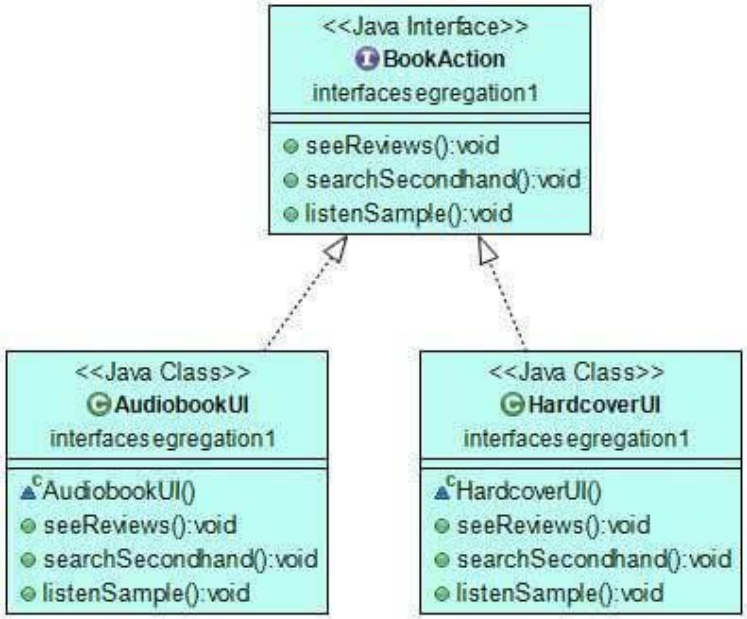
```java
class HardcoverUI implements HardcoverAction {

    @Override
    public void seeReviews() {...}

    @Override
    public void searchSecondhand() {...}

}

class AudiobookUI implements AudioAction {

    @Override
    public void seeReviews() {...}

    @Override
    public void listenSample() {...}

}
```

Now, the **HardcoverUI** class can implement the ***HardcoverAction*** interface and the **AudiobookUI** class can implement the ***AudioAction*** interface.

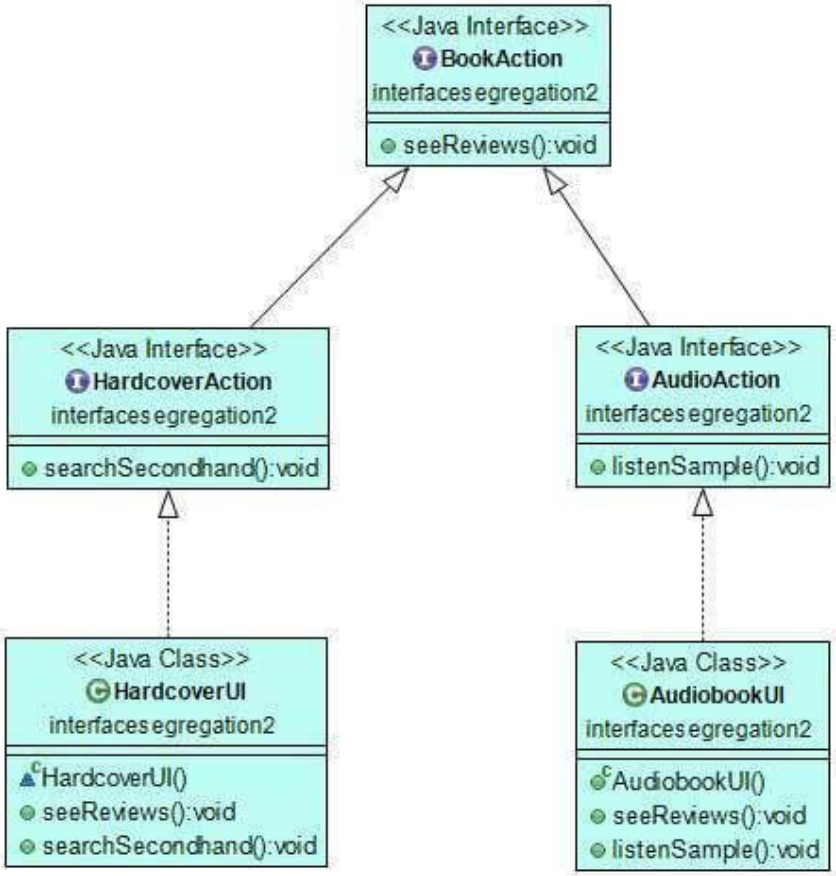
This way, both classes can implement the seeReviews() method of the **BookAction** super-interface. However, **HardcoverUI** doesn't have to implement the **irrelevant** listenSample() method and **AudioUI** doesn't have to implement searchSecondhand(), either.

Interface Segregation Principle

Before refactoring



After refactoring



Interface Segregation Principle

- A client should never be forced to depend on methods it doesn't use.
- Or, a client should never depend on anything more than the method it's calling.
- Changing one method in a class shouldn't affect classes that don't depend on it.
- Replace fat interfaces with many small, specific interfaces

Interface Segregation Principle

- The *Interface Segregation Principle* (ISP) states that the client should never be forced to depend on an interface they aren't using in its entirety. This means that an interface should have a **minimum set of methods** necessary for the functionality it ensures, and should be limited to only one functionality.

Dependency Inversion Principle

- The goal of the **Dependency Inversion Principle** is to avoid tightly coupled code, as it easily breaks the application. The principle states that:
  - *"High-level modules should not depend on low-level modules. Both should depend on abstractions."*
  - *"Abstractions should not depend on details. Details should depend on abstractions."*
- In other words, we need to decouple high-level and low-level classes. High-level classes usually encapsulate complex logic while low-level classes include data or utilities. Typically, most people would want to make high-level classes depend on low-level classes. However, according to the Dependency Inversion Principle, we need to invert the dependency. Otherwise, when the low-level class is replaced, the high-level class will be affected, too.
- As a solution, you need to create an abstract layer for low-level classes, so that high-level classes can depend on abstraction rather than concrete implementations.
- Dependency Inversion Principle is a specific combination of the Open/Closed and Liskov Substitution Principles.

Dependency Inversion Principle

- Now, the book store requires to add a new feature that enables customers to put their favorite books on a shelf.
- To implement the new functionality, we create a **lower-level Book class and a higher-level Shelf class**.

```
class Book {

    void seeReviews() {...}
    void readSample() {...}

}

class Shelf {

    Book book;

    void addBook(Book book) {...}
    void customizeShelf() {...}

}
```

- Next, the store requires to enable customers to add DVDs to their shelves, too.
- To fulfill the demand, we create a **new DVD class**:

```
class DVD {

    void seeReviews() {...}
    void watchSample() {...}

}
```

- and modify the **Shelf class** so that it can accept **DVDs**, too.

```
class Shelf {

    Book book;

    DVD dvd;

    void addBook(Book book) {...}
    void addDVD(DVD dvd) {...}
    void customizeShelf() {...}

}
```

This clearly breaks the Open/Closed Principle

Dependency Inversion Principle

The **solution** is to create an abstraction layer for the lower-level classes (Book and DVD) by introducing the **Product interface** that both classes will implement.

Now, Shelf can reference the Product interface instead of its implementations (Book and DVD).

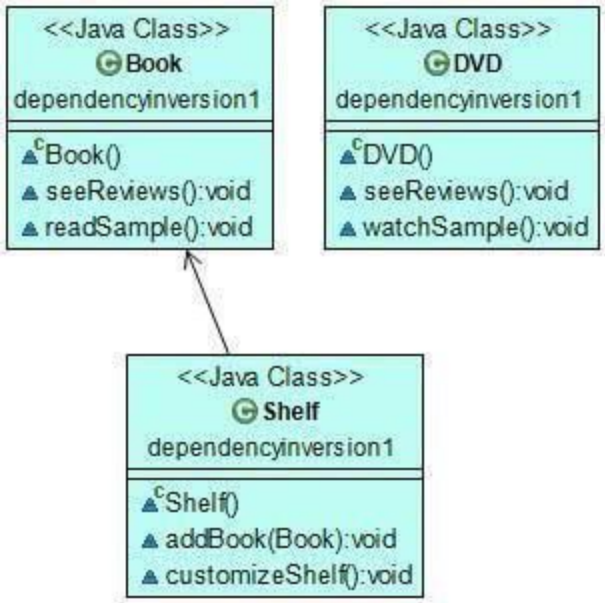
```
class Shelf {

    Product product;

    void addProduct(Product product) {...}

    void customizeShelf() {...}

}
```

```
public interface Product {

    void seeReviews();
    void getSample();

}

class Book implements Product {

    @Override
    public void seeReviews() {...}

    @Override
    public void getSample() {...}

}

class DVD implements Product {

    @Override
    public void seeReviews() {...}

    @Override
    public void getSample() {...}

}
```

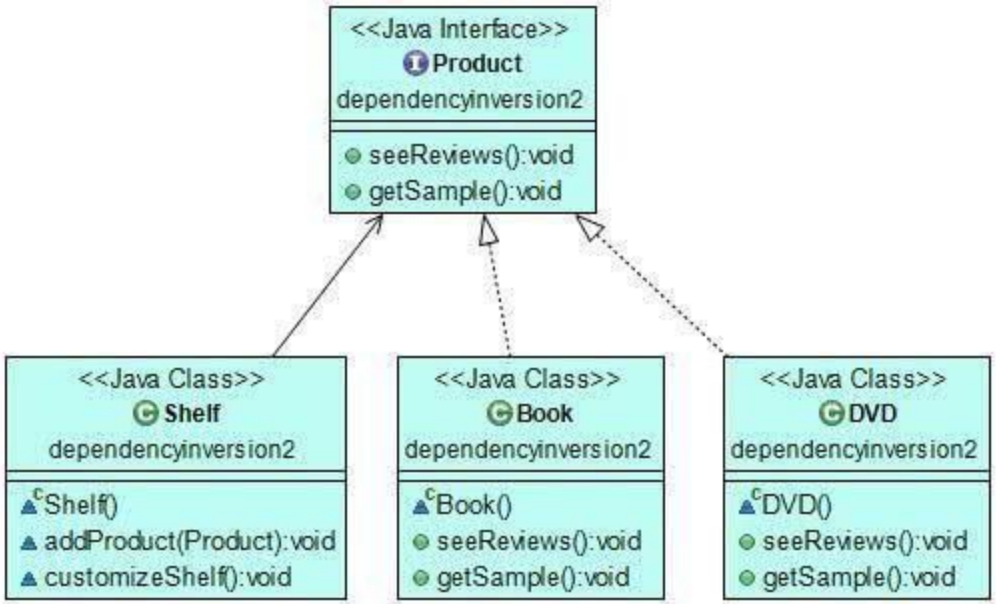
The above code also follows the Liskov Substitution Principle, as the Product type can be substituted with both of its subtypes (Book and DVD) without breaking the program. At the same time, it also implemented the Dependency Inversion Principle, as in the refactored code, high-level classes don't depend on low-level classes, either.

Dependency Inversion Principle

Before refactoring



After refactoring



Dependency Inversion Principle

- Never depend on anything concrete, only depend on abstractions.
- High level modules should not depend on low level modules. They should depend on abstractions.
- Able to change an implementation easily without altering the high level code.

Dependency Inversion Principle

- If Dependency Inversion is not implemented in the code, we run the risk of:
  - Damaging the higher level code that uses the lower level classes.
  - Requiring too much time and effort to change the higher level code when a change occurs in the lower level classes.
  - Producing less-reusable code.

Dependency Inversion Principle

- According to the Dependency Inversion Principle (DIP), high-level and low-level modules should be decoupled in such a way that changing (or even replacing) low-level modules doesn't require (much) rework of high-level modules. Given that, both low-level and high-level modules shouldn't depend on each other, but rather they should depend on abstractions, such as interfaces.
- Another important thing **DIP** states is:
  - Abstractions should not depend on details. Details (concrete implementations) should depend on abstractions.
- This principle is important because it decouples modules, making the system less complex, easier to maintain and update, easier to test, and more reusable. I can't stress enough how much of a game changer this is, especially for unit testing and reusability. If the code is written generically enough, it can easily find application in another project, while code that's too specific and interdependent with other modules of the original project will be hard to decouple from it.

Don’t get trapped by SOLID

- SOLID design principles are principles, not rules.
- Always use common sense when applying SOLID.
- Avoid over-fragmenting your code for the sake of SRP of SOLID.
- Don’t try to achieve SOLID, use SOLID to achieve maintainability.